consider $\dot{x} = \begin{bmatrix} a_1 & a_2 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$, for $x \in \mathbb{R}$, $u \in \mathbb{R}$

$J = \frac{1}{2}\int_0^{\infty} c_{11}x_1^2 + c_{12}x_1x_2 + c_{22}x_2 + u^2 \, dt = \frac{1}{2}\int_0^{\infty} q(x) + u^2 \, dt$, $q(x)$ is positive definite

$$H(x, u, \lambda) = \frac{1}{2}q(x) + \frac{1}{2}u^2 + \lambda^T \dot{x}$$

$$= \frac{1}{2}q(x) + \frac{1}{2}u^2 + \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}^T \left( \begin{bmatrix} a_1 & a_2 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \right)$$

$$= \frac{1}{2}q(x) + \frac{1}{2}u^2 + [\lambda_1 \;\; \lambda_2] \begin{bmatrix} a_1x_1 + a_2x_2 + u \\ -x_2 \end{bmatrix}$$

$$= \frac{1}{2}(c_{11}x_1^2 + c_{12}x_1x_2 + c_{22}x_2) + \frac{1}{2}u^2 + \lambda_1(a_1x_1 + a_2x_2 + u) - \lambda_2x_2$$

$$\therefore \frac{\partial H}{\partial u} = u + \lambda_1 \qquad \therefore \frac{\partial^2 H}{\partial u^2} = 1 > 0$$

$$\therefore \left.\frac{\partial H}{\partial u}\right|_{u=u^*} = u^* + \lambda_1 = 0$$

$$\therefore u^* = -\lambda_1$$

$$\therefore H^*(x, \lambda) = \frac{1}{2}(c_{11}x_1^2 + c_{12}x_1x_2 + c_{22}x_2) + \lambda_1(a_1x_1 + a_2x_2 - \frac{1}{2}\lambda_1) - \lambda_2x_2 = 0$$

suppose $a_1 = -2$, $a_2 = 2$, find a set of values $c_{11}, c_{12}, c_{22}$ such that $u^* = -x_1$

$$\because \lambda = \frac{\partial V}{\partial x} \text{ in Dynamic Programming}$$

$$\therefore u^* = -\frac{\partial V}{\partial x_1} = -x_1$$

$$\therefore V = \frac{1}{2}x_1^2 + f(x_2)$$

$$\therefore \frac{1}{2}(c_{11}x_1^2 + c_{12}x_1x_2 + c_{22}x_2) + x_1(-2x_1 + 2x_2 - \frac{1}{2}x_1) - \frac{df(x_2)}{dx_2}x_2 = 0$$

$$\therefore \begin{cases} (\frac{1}{2}c_{11} - \frac{5}{2})x_1^2 = 0 \\ (\frac{1}{2}c_{12} + 2)x_1 x_2 = 0 \\ \frac{1}{2}c_{22}x_2^2 - \frac{df(x_2)}{dx_2} = 0 \end{cases}$$

$$\therefore \begin{cases} c_{11} = 5 \\ c_{12} = -4 \\ f(x_2) = \int \frac{1}{2}c_{22}x_2 \, dx_2 = \frac{1}{4}c_{22}x_2^2 + C \end{cases} \quad \because f(0) = 0 \text{ since } V(x, 0) = 0 \quad \therefore C = 0$$

any selection of $c_{22}$ for $q(x) > 0$

$$\therefore \frac{1}{2}x^T \begin{bmatrix} 5 & -2 \\ -2 & c_{22} \end{bmatrix} x > 0$$

$$\therefore \begin{bmatrix} 5 & -2 \\ -2 & c_{22} \end{bmatrix} > 0 \quad (\text{positive definite})$$

what is value function corresponding to arbitory $c_{22}$ satisfy above constraint ?

$$V = \frac{1}{2}x_1^2 + f(x_2) = \frac{1}{2}x_1^2 + \frac{1}{2}c_{22}x_2^2$$

$$= \frac{1}{2}x^T \begin{bmatrix} 1 & 0 \\ 0 & c_{22} \end{bmatrix} x = \frac{1}{2}x^T P x$$

* can verify that $P = P^T > 0$ satisfies the ARE

$$PA + A^T P - PBR^{-1}BP + Q = 0$$